

CSE461 - Practice Question Bank with Answers [summer26_test]

CSE461 | Practice Question Bank

CSE461 | Practice Question Bank | Page 1 Part A: MCQ (50 questions)

1. A robot arm in a car factory repeatedly picks engine parts from a conveyor and places them into a fixture. The most important performance property for this task is:

- A) Repeatability so the arm returns to the same position every cycle**
- B) Maximum speed regardless of accuracy
- C) The ability to learn new tasks every hour
- D) Running without any power source

Answer: A) Repeatability so the arm returns to the same position every cycle

A pick-and-place arm must return to the same position every cycle. Repeatability is the defining property for this task.

2. A warehouse mobile robot is navigating and a worker suddenly steps into its path. The part of the robot system that responds immediately without waiting for a new plan is:

- A) The global path planner
- B) The map builder
- C) The reactive layer**
- D) The battery management unit

Answer: C) The reactive layer

The reactive layer responds immediately to sensor input without re-planning. This is its core purpose.

3. A robot is designed to detect gas leaks inside chemical storage tanks. The best primary justification from the Four Ds for using a robot here is:

- A) Dull because gas detection is a boring task
- B) Dirty because tanks are covered in grime
- C) Difficult because gas molecules are hard to see
- D) Dangerous because toxic gas is immediately harmful to humans**

Answer: D) Dangerous because toxic gas is immediately harmful to humans

Toxic environments that threaten human life are the strongest Dangerous case in the Four Ds framework.

4. Which of the following best describes what a sensor does in a robotic system?

- A) It generates the movement commands for the motors
- B) It stores the robot navigation map permanently
- C) It detects a physical property of the environment and converts it into a usable signal**
- D) It manages the power supply to all actuators

Answer: C) It detects a physical property of the environment and converts it into a usable signal

A sensor detects a physical property such as distance temperature or light and converts it into a usable signal.

5. A cleaning robot navigates a hospital corridor at night with all lights off. Which sensor is least affected by the complete absence of light?

- A) A standard RGB camera
- B) An infrared proximity sensor that relies on ambient light reflection
- C) A LiDAR sensor**
- D) A colour detection camera

Answer: C) A LiDAR sensor

LiDAR emits its own laser pulses and does not depend on ambient light. It works in complete darkness.

6. A delivery robot uses a global map of a building to plan its route from the loading bay to office

304. The part of the robot system responsible for this planning is:

- A) The reactive obstacle avoidance layer
- B) The pneumatic actuator
- C) The wheel encoder
- D) The deliberative planning layer**

Answer: D) The deliberative planning layer

The deliberative or planning layer uses the global map to compute the route to the destination.

7. A robot navigates a room full of soft foam furniture using an ultrasonic sensor. Compared to hard wooden furniture at the same distance the foam will:

- A) Return a stronger echo because foam is denser than wood
- B) Return a weaker echo or cause missed readings because foam absorbs sound energy**
- C) Make the sensor read a shorter distance than reality
- D) Have no effect because ultrasonic sensors use light not sound

Answer: B) Return a weaker echo or cause missed readings because foam absorbs sound energy

Foam absorbs acoustic energy rather than reflecting it cleanly back to the receiver causing weak or absent return signals.

8. A robot gripper must hold a fragile ceramic bowl without breaking it. The best actuator choice for the gripper fingers is:

- A) Hydraulic cylinder because it produces maximum force
- B) Servo motor with force feedback because it can regulate grip force precisely**
- C) Stepper motor at full power because it holds position firmly
- D) DC motor without feedback because it is the simplest option

Answer: B) Servo motor with force feedback because it can regulate grip force precisely

A servo motor with force feedback can apply and limit grip force to prevent overforce on fragile objects.

9. Which control paradigm is used in a robot that has no memory no map and makes every decision purely from what its sensors are reading at that exact moment?

- A) Hierarchical paradigm
- B) Hybrid paradigm
- C) Reactive paradigm**
- D) Deliberative paradigm

Answer: C) Reactive paradigm

The reactive paradigm operates with no memory or map. Every decision comes entirely from immediate sensor readings.

10. A robot uses a gear system where the output gear has more teeth than the input gear. What happens to the output shaft speed compared to the input shaft speed?

- A) The output shaft spins faster than the input shaft
- B) The output shaft spins at the same speed as the input shaft
- C) The output shaft stops completely when tooth count increases
- D) The output shaft spins slower than the input shaft and torque increases**

Answer: D) The output shaft spins slower than the input shaft and torque increases

When the output gear has more teeth it rotates slower. Torque increases proportionally as speed decreases. This is the fundamental principle of gear reduction.

11. A robot navigates a narrow aisle and its ultrasonic sensor faces a smooth glass panel angled slightly away from the beam. The sensor will most likely:

- A) Detect the glass wall perfectly because glass is a solid object
- B) Miss the glass wall because the angled smooth surface reflects sound away from the receiver**
- C) Report the glass wall at double the real distance
- D) Switch to a camera automatically when glass is detected

Answer: B) Miss the glass wall because the angled smooth surface reflects sound away from the receiver

Smooth angled surfaces deflect the ultrasonic beam away from the receiver rather than reflecting it back causing a missed detection.

12. A robot operating outdoors in dense fog uses LiDAR for navigation. The most significant problem caused by the fog is:

- A) The LiDAR motor becomes wet and stops spinning
- B) Water droplets in the fog reflect laser pulses and create many false obstacle readings throughout the scan**
- C) Fog causes the LiDAR to scan at a slower rate
- D) LiDAR only works in direct sunlight so fog blocks all readings

Answer: B) Water droplets in the fog reflect laser pulses and create many false obstacle readings throughout the

Each water droplet can return a laser pulse filling the scan with false positives that corrupt the obstacle map.

13. A robot must pick ripe tomatoes from plants in a greenhouse and distinguish them from unripe ones and green leaves. The most appropriate sensor for this colour discrimination is:

- A) Ultrasonic sensor
- B) Wheel encoder
- C) RGB camera with colour processing**
- D) Infrared proximity sensor

Answer: C) RGB camera with colour processing

Colour discrimination between red and green objects requires a camera with colour image processing capability.

14. A robot arm joint uses a large output gear driven by a small input gear. Compared to using gears of equal size this configuration will:

- A) Increase output speed and decrease output torque
- B) Only change direction of rotation without affecting speed or torque
- C) Keep speed and torque identical to equal-sized gears
- D) Decrease output speed and increase output torque**

Answer: D) Decrease output speed and increase output torque

A larger output gear reduces speed and multiplies torque. This is the fundamental principle of gear reduction.

15. A mobile robot in a library must navigate narrow aisles between bookshelves and turn sharply without hitting shelves. The most suitable drive configuration is:

- A) Differential drive because it can rotate in place by driving wheels in opposite directions**
- B) Four-wheel drive with fixed axles because it provides more traction
- C) Single front-wheel steering because it is mechanically simple
- D) Hydraulic drive because it provides higher torque

Answer: A) Differential drive because it can rotate in place by driving wheels in opposite directions

Differential drive allows the robot to spin in place by reversing one wheel making it ideal for very tight spaces.

16. A robot uses an infrared sensor to detect a table edge. The table surface is polished black marble. Compared to a white table surface the sensor will:

- A) Detect the edge at greater distance because black reflects more energy
- B) Work identically because table edge geometry is the same
- C) Underperform because dark surfaces absorb infrared and return very little signal back to the receiver**
- D) Switch to ultrasonic mode automatically

Answer: C) Underperform because dark surfaces absorb infrared and return very little signal back to the receiver

Dark surfaces absorb IR radiation. Very little reflected energy reaches the sensor receiver making detection unreliable on dark surfaces.

17. A purely hierarchical robot would struggle to work safely alongside humans in a busy office because:

- A) The full sense-plan-act cycle takes time making the robot too slow to react to unpredictable human movement**
- B) Hierarchical robots are too fast for humans to follow
- C) Hierarchical robots cannot store maps
- D) Hierarchical robots only work outdoors

Answer: A) The full sense-plan-act cycle takes time making the robot too slow to react to unpredictable human

The full sense-plan-act cycle introduces delay. In environments where humans move unpredictably this delay makes hierarchical-only control unsafe.

18. A robot patrol security guard navigates a museum at night. It needs to detect a human body that is completely still in a dark corner. Which sensor is most suited to this specific task?

- A) RGB camera because it provides detailed images
- B) Ultrasonic sensor because it detects solid objects
- C) Thermal infrared camera because it detects body heat regardless of light level**
- D) LiDAR because it scans in 360 degrees

Answer: C) Thermal infrared camera because it detects body heat regardless of light level

A thermal camera detects heat emitted by the human body. It works in complete darkness and can find a motionless person that other sensors might miss.

19. A robot arm joint uses a gear drive where the gear ratio and friction cause the output to self-lock when the motor is off. A heavy load is attached. When the motor is switched off the arm holds its position. This self-locking behaviour occurs because:

- A) The motor creates a magnetic lock when powered off
- B) The gear geometry and friction at the contact surface prevent the load from back-driving the input shaft**
- C) The output gear is attached to a spring that resists gravity
- D) All gear drives automatically lock when the motor stops

Answer: B) The gear geometry and friction at the contact surface prevent the load from back-driving the input shaft

Self-locking occurs when back-driving from the output side would require more force than friction and geometry allow. The load cannot rotate back through the gear.

20. A robot arm performs spot welding on car bodies and must reach the same joint angle exactly at every cycle throughout a full factory shift. Which motor is most appropriate for each joint?

- A) Servo motor with encoder feedback because it actively corrects position error every cycle**
- B) DC motor without feedback because it is low cost
- C) Stepper motor because it moves in fixed increments
- D) Pneumatic motor because it is fast

Answer: A) Servo motor with encoder feedback because it actively corrects position error every cycle

Servo motors with encoders continuously correct position error maintaining accuracy across thousands of repetitions even as load and temperature vary.

21. A robot exploring a collapsed building after an earthquake has no pre-built map. It must build a map while navigating and avoid falling debris at the same time. The most appropriate control paradigm is:

- A) Purely hierarchical because it plans the best route in advance
- B) Purely reactive because it responds immediately to every obstacle
- C) No paradigm is needed because the robot only uses GPS
- D) Hybrid combining simultaneous mapping with reactive local obstacle avoidance**

Answer: D) Hybrid combining simultaneous mapping with reactive local obstacle avoidance

The hybrid paradigm allows the robot to build a map incrementally while the reactive layer handles immediate obstacles in an environment that cannot be pre-mapped.

22. A chain drive is preferred over a belt drive in a robot that must operate in a food cold storage room at minus 10 degrees Celsius. The reason is:

- A) Chains are quieter in cold conditions
- B) Rubber belts harden and crack in very cold temperatures while metal chains remain functional**
- C) Chains produce less friction heat which is important in cold rooms
- D) Chains are lighter than belts at low temperatures

Answer: B) Rubber belts harden and crack in very cold temperatures while metal chains remain functional

Cold temperatures cause rubber to lose flexibility and crack. Metal chains are unaffected by typical cold storage temperatures and continue operating reliably.

23. A robot must move a camera platform smoothly and quietly along a horizontal rail in a recording studio. Which transmission is most appropriate?

- A) Roller chain because it is strong and durable
- B) Synchronous belt drive because it is quiet smooth and suited to low-noise environments**
- C) A large gear reduction stage because it provides high torque
- D) A rack and pinion using a large gear because it is stiff

Answer: B) Synchronous belt drive because it is quiet smooth and suited to low-noise environments

Synchronous belt drives are quiet and smooth making them ideal for noise-sensitive environments like recording studios.

24. A mobile robot uses only wheel odometry to navigate across a wet floor. After some distance its position estimate is significantly wrong. The most likely cause is:

- A) Wheel encoders overheat on wet floors
- B) Water blocks the encoder signal
- C) Wet floors make the robot heavier slowing the encoder
- D) Wheel slipping on the wet surface causes the encoder to count rotations that did not result in actual forward motion**

Answer: D) Wheel slipping on the wet surface causes the encoder to count rotations that did not result in actual

When wheels slip they rotate without the robot actually moving forward by the expected amount. The encoder counts those rotations as distance creating position error.

25. A rack-and-pinion mechanism is used on a robot to extend a linear probe. The main advantage over a belt for this linear task is:

- A) It is quieter than a belt drive
- B) It provides stiffer and more precise linear motion with no stretch or compliance**
- C) It works without any motor
- D) It can route around corners easily

Answer: B) It provides stiffer and more precise linear motion with no stretch or compliance

Rack-and-pinion is mechanically rigid. Unlike belts there is no stretch or elasticity giving more accurate and repeatable linear positioning.

26. A robot must carry a 500 kg load of concrete debris on a construction site. Electric servo motors are proposed. A better choice for this specific heavy load task would be:

- A) Stepper motors because they hold position without power
- B) A smaller servo motor because it is more precise
- C) Hydraulic actuators because they produce far higher forces than electric motors of comparable size**
- D) Pneumatic actuators because compressed air is available on construction sites

Answer: C) Hydraulic actuators because they produce far higher forces than electric motors of comparable size

Hydraulic actuators have very high force density and can produce much larger forces than comparably sized electric motors making them appropriate for very heavy loads.

27. A robot is navigating through a greenhouse and the path planning layer has computed a route between plant rows. Midway through a plant pot has been moved into the path. The correct system response in a hybrid paradigm is:

- A) Stop permanently and wait for a human to move the pot
- B) Cancel the task and return to the start
- C) Re-plan the entire route from the beginning before moving at all
- D) The reactive layer avoids the pot locally while the planning layer continues directing the robot toward the goal**

Answer: D) The reactive layer avoids the pot locally while the planning layer continues directing the robot toward the

In a hybrid paradigm the reactive layer handles local unexpected obstacles while the planning layer maintains the overall goal. The robot avoids the pot and continues.

28. A servo motor on a robot arm receives a position command but an external load pushes the joint slightly off target. What happens next?

- A) **The encoder detects the position error and the controller drives the motor back to the target position**
- B) The servo stops permanently because it cannot handle the error
- C) The motor ignores the error because position commands override sensor feedback
- D) The motor reverses direction and moves away from the target

Answer: A) The encoder detects the position error and the controller drives the motor back to the target position

Servo motors continuously compare actual position from the encoder with the commanded position and apply corrective drive to eliminate any error.

29. A stepper motor positions a valve in a water treatment robot. The valve is stuck due to rust and the motor cannot move it. The controller continues sending pulses. What is the most likely outcome?

- A) The motor detects the resistance and applies more torque automatically
- B) **The motor skips steps silently and the controller believes the valve is at the commanded position when it is not**
- C) The motor shuts down automatically to protect itself
- D) The motor sends a fault signal to the controller immediately

Answer: B) The motor skips steps silently and the controller believes the valve is at the commanded position when it

Stepper motors have no position feedback. If load prevents movement pulses are skipped silently and the controller loses track of true position with no indication of error.

30. A PLC and a Raspberry Pi are both proposed for controlling a packaging line robot running 24 hours per day. The Raspberry Pi is rejected mainly because:

- A) Raspberry Pi cannot run any real-time tasks at all
- B) Raspberry Pi is too expensive for industrial use
- C) **A general-purpose Linux system can have unpredictable delays and restarts that are unacceptable in continuous industrial control**
- D) Raspberry Pi does not have any output ports for motors

Answer: C) A general-purpose Linux system can have unpredictable delays and restarts that are unacceptable in

Linux-based SBCs can experience OS-level scheduling delays and unplanned reboots. PLCs execute control cycles deterministically with no such interruptions.

31. A Jetson Nano is selected over an Arduino for a robot that performs real-time object detection from a camera. The main reason is:

- A) **Jetson Nano has a GPU that can process image data at high speed using neural networks**
- B) Jetson Nano uses less power than Arduino
- C) Jetson Nano is physically smaller than Arduino
- D) Jetson Nano does not need any programming

Answer: A) Jetson Nano has a GPU that can process image data at high speed using neural networks

The Jetson Nano has a GPU designed for neural network inference enabling real-time object detection that is impossible on an Arduino.

32. A robot is used to paint the inside of large chemical storage tanks. The primary justification from the Four Ds is:

- A) Difficult because painting is technically complex
- B) Dull because painting is repetitive
- C) Dirty and Dangerous because the tank interior is chemically hazardous and confined**
- D) Difficult and Dull because humans find painting boring and complicated

Answer: C) Dirty and Dangerous because the tank interior is chemically hazardous and confined

Chemical tank interiors contain toxic fumes and confined space dangers. Dirty and Dangerous categories both strongly apply.

33. Asimov's First Law states that a robot must not harm a human. A robot surgeon receives a command to perform an incision deeper than is medically safe for the patient. According to the First Law the robot should:

- A) Refuse because following the command would harm the patient**
- B) Follow the command because the surgeon has authority
- C) Ask for a second surgeon's opinion before proceeding
- D) Perform the incision at half depth as a compromise

Answer: A) Refuse because following the command would harm the patient

The First Law overrides all human instructions when following them would cause harm to a human being.

34. A LiDAR sensor scans a room and points at a large mirror on the wall. What will the LiDAR most likely report?

- A) A wall at the correct mirror distance
- B) Objects or walls reflected in the mirror appearing as if they exist behind the mirror**
- C) An extremely bright return signal indicating glass
- D) An error code because mirrors are not a supported surface type

Answer: B) Objects or walls reflected in the mirror appearing as if they exist behind the mirror

A mirror specularly reflects the laser beam. The LiDAR measures the full reflected path and may detect the reflected scene as ghost readings of objects not physically present.

35. A robot navigates a corridor and both its left and right ultrasonic sensors detect walls simultaneously. The reactive obstacle avoidance pushes the robot away from both walls equally. What is the likely result?

- A) The robot accelerates forward because both forces cancel
- B) The robot turns right because right has priority
- C) The robot freezes or oscillates in the corridor centre unable to progress**
- D) The robot increases sensor range to find a wider path

Answer: C) The robot freezes or oscillates in the corridor centre unable to progress

When equal repulsive forces from both sides cancel each other the robot can enter a local minimum where it cannot move forward. This is a known failure of purely reactive systems.

36. A microcontroller handles motor PWM control and encoder reading while an SBC handles camera processing and path planning on the same robot. This division is correct because:

- A) Microcontrollers handle low-level real-time tasks well while SBCs handle computation-heavy high-level tasks**
- B) Microcontrollers are cheaper so they should do more work
- C) SBCs cannot send PWM signals to motors under any circumstances
- D) This division is done only to reduce cable length between components

Answer: A) Microcontrollers handle low-level real-time tasks well while SBCs handle computation-heavy high-level

Microcontrollers excel at deterministic low-level timing tasks. SBCs have the processing power for camera processing and planning algorithms.

37. A robot designed to search for survivors in a dark underground cave has no GPS signal no ambient light and uneven rocky terrain. The sensing approach most suited to navigation here is:

- A) RGB camera because it provides the richest visual information
- B) GPS and compass because they give absolute position
- C) LiDAR combined with IMU for mapping and pose estimation in GPS-denied dark environments**
- D) Single infrared proximity sensor because caves are dark

Answer: C) LiDAR combined with IMU for mapping and pose estimation in GPS-denied dark environments

LiDAR does not need light and can map 3D geometry. Combined with an IMU for motion estimation it enables navigation in GPS-denied dark environments.

38. A robot arm joint uses a belt drive that begins to slip under high load. The immediate consequence for the joint is:

- A) The joint moves faster because belt tension decreases
- B) The motor draws higher current and shuts down safely
- C) The belt becomes louder but continues delivering full torque
- D) The joint falls behind the commanded position because slipping transfers less force to the output**

Answer: D) The joint falls behind the commanded position because slipping transfers less force to the output

Belt slip means the input and output are no longer synchronised. Less torque reaches the joint and it falls behind the commanded position.

39. A robot floor sensor uses infrared to detect surface changes. The robot moves from a white tile floor to a dark wood floor. What will the sensor experience at the boundary?

- A) No change because floor colour does not affect infrared sensors
- B) A sudden drop in returned signal strength because dark wood absorbs more infrared than white tile**
- C) The robot stops because dark floors are always detected as drop-offs
- D) The sensor switches automatically to ultrasonic mode at the boundary

Answer: B) A sudden drop in returned signal strength because dark wood absorbs more infrared than white tile

Dark wood absorbs more IR than white tile. The returned signal drops sharply at the colour boundary which the system interprets as a surface change.

40. A robot working in a paint spray booth uses pneumatic actuators. The engineer discovers that exhausted air from the actuators disrupts the spray pattern and causes uneven paint quality. The best solution is:

- A) Increase air pressure to make the spray stronger
- B) Replace the pneumatic actuators with electric actuators that do not exhaust air into the booth environment**
- C) Add more spray nozzles to compensate for the disruption
- D) Move the actuators outside the booth and use very long linkage arms

Answer: B) Replace the pneumatic actuators with electric actuators that do not exhaust air into the booth

Electric actuators do not exhaust air eliminating the airflow disruption that affects paint quality inside the spray booth.

41. A robot tries to localise itself in a long straight corridor where all doors look identical. The robot cannot determine which door it is in front of even after moving several metres. This problem is best described as:

- A) Sensor saturation
- B) Perceptual aliasing where different locations produce indistinguishable sensor readings**
- C) Map inflation
- D) Odometric divergence

Answer: B) Perceptual aliasing where different locations produce indistinguishable sensor readings

Perceptual aliasing occurs when distinct locations produce identical sensor observations making unambiguous localisation impossible.

42. A robot must apply a label to a product moving on a conveyor and must complete this action within a very tight fixed time window. The control hardware must guarantee this timing with no variation. Which is most appropriate?

- A) Raspberry Pi running Linux because it is fast
- B) Cloud computing because it has unlimited resources
- C) Arduino because it is simple to program
- D) PLC because it executes control cycles with deterministic timing guarantees**

Answer: D) PLC because it executes control cycles with deterministic timing guarantees

PLCs execute in deterministic cycles with guaranteed response times. Linux systems and cloud connections cannot guarantee the fixed timing required.

43. A robot joint driven by a servo motor receives a command to move to a target position. An obstacle prevents the joint from reaching the target. The servo motor will:

- A) Accept the blocked position as the new target automatically
- B) Continuously try to reach the target applying force until it reaches the position or a torque limit is hit**
- C) Shut down immediately to prevent damage
- D) Skip to the next command in the sequence

Answer: B) Continuously try to reach the target applying force until it reaches the position or a torque limit is hit

The servo controller continuously compares target and actual position. If blocked it keeps applying drive force to reach the target until a torque limit or timeout is reached.

44. A robot uses only a camera for obstacle detection and enters a room where the floor and walls are all painted the same flat grey colour with no texture. The most likely navigation problem is:

- A) The camera overheats in grey environments
- B) The robot cannot distinguish the floor from the walls due to lack of visual contrast causing navigation failure**
- C) Grey colour permanently damages the camera image sensor
- D) The camera automatically switches to depth sensing mode in low-contrast environments

Answer: B) The robot cannot distinguish the floor from the walls due to lack of visual contrast causing navigation

Camera-based navigation relies on visual contrast to detect surface boundaries. Uniform colour and texture provide no features for segmentation or localisation.

45. A rack-and-pinion system extends a robotic arm along a linear track. If the engineer replaces the current pinion with a smaller one while keeping the same rack the effect will be:

- A) The arm moves further per motor revolution
- B) The arm moves less distance per motor revolution because the smaller pinion has a smaller circumference**
- C) The arm moves at the same speed because rack tooth spacing does not change
- D) The arm reverses direction because the pinion is smaller

Answer: B) The arm moves less distance per motor revolution because the smaller pinion has a smaller

A smaller pinion has a smaller pitch circumference. Each revolution moves the rack a shorter distance while torque at the rack increases.

46. A robot designed to transport medicine carts along fixed hospital corridors for eight hours per day is proposed. The best argument that this qualifies for the Dull category of the Four Ds is:

- A) The medicine carts are heavy and hard to push
- B) The corridors are dirty from foot traffic
- C) The task is repetitive and monotonous causing human fatigue and reduced reliability over long shifts**
- D) Hospital corridors are dangerous because of sick patients nearby

Answer: C) The task is repetitive and monotonous causing human fatigue and reduced reliability over long shifts

Dull tasks are repetitive and fatiguing over long shifts. Human attention and consistency degrade making automation beneficial.

47. A stepper motor loses track of its true position because the load caused it to skip steps during a fast move. The only reliable way to restore correct position knowledge is:

- A) Sending extra pulses in the same direction to compensate
- B) Running a homing routine that moves the joint to a known physical reference and resets the position counter from there**
- C) Turning the power off and on to reset the controller
- D) Increasing stepping frequency so the motor catches up automatically

Answer: B) Running a homing routine that moves the joint to a known physical reference and resets the position

Without feedback the only way to recover known position after missed steps is to move to a physical reference such as a limit switch and reset from that known state.

48. An engineer is choosing between a belt drive and a chain drive for a robot operating in an outdoor agricultural field with mud water and direct sunlight. The better choice is:

- A) Belt drive because belts are more flexible and handle terrain better
- B) Chain drive because metal resists mud moisture and UV exposure better than rubber**
- C) Belt drive because it is lighter and reduces power consumption
- D) Chain drive because it is always quieter outdoors

Answer: B) Chain drive because metal resists mud moisture and UV exposure better than rubber

Metal chain drives resist moisture UV and contamination better than rubber belts which can swell crack or slip when exposed to outdoor conditions.

49. A student builds a reactive robot to escape a room through a door placed in a corner. The robot uses proximity sensors and moves away from detected objects. The most likely failure is:

- A) The robot finds the door faster because corners have strong sensor signals
- B) The robot gets trapped in the corner because repulsive forces from two walls create a local minimum**
- C) The robot exits faster because it has two references to navigate toward
- D) The robot improves its internal map and routes around the corner

Answer: B) The robot gets trapped in the corner because repulsive forces from two walls create a local minimum

A corner creates two simultaneous repulsive forces. If these balance before the robot reaches the door it settles in the corner equilibrium and cannot escape.

50. A gear system on a robot changes the direction of rotation from one shaft to another shaft oriented at 90 degrees to it. What is the main purpose of this type of gear arrangement?

- A) To increase the speed of the output shaft
- B) To lock the output shaft when the motor is off
- C) To convert rotary motion into linear motion like a rack and pinion
- D) To transmit motion between two shafts that are not parallel**

Answer: D) To transmit motion between two shafts that are not parallel

Certain gear arrangements transmit rotary motion between non-parallel shafts redirecting the axis of rotation. This is the core purpose of right-angle gear drives.

Part B: True / False (50 questions)

1. A purely reactive robot cannot navigate to a specific destination across a large building without any form of goal representation or planning.

Answer: True

Reactive systems respond only to current sensor input. Without a goal or planning layer the robot has no mechanism to guide itself toward a specific distant target.

2. Asimov's Second Law allows a robot to follow a human command even when doing so would harm another human being.

Answer: False

The Second Law requires obedience only when it does not conflict with the First Law. If a command causes harm the First Law takes priority and the robot must refuse.

3. A servo motor continuously measures its actual position and uses that measurement to correct any difference from the commanded position.

Answer: True

This closed-loop feedback is the defining feature of a servo motor. The controller corrects position error throughout the motion.

4. A chain drive is more suitable than a rubber belt drive in environments where temperature regularly drops below minus 10 degrees Celsius.

Answer: True

Rubber hardens and cracks at low temperatures. Metal chains retain their mechanical properties across a much wider temperature range.

5. An ultrasonic sensor will reliably detect a large pane of glass placed at a 45-degree angle to the sensor beam.

Answer: False

Glass at an angle reflects sound away from the receiver rather than back toward it. The sensor misses the glass or returns a very weak signal.

6. The deliberative planning layer in a hybrid robot is responsible for computing the overall route or task sequence.

Answer: True

The deliberative layer uses a world model and goals to plan the path or action sequence the robot should follow.

7. A camera-based navigation system works reliably in a tunnel where all walls are painted the same uniform colour with no visible texture.

Answer: False

Camera navigation depends on visual features and contrast. Uniform colour with no texture provides no landmarks for localisation or obstacle detection.

8. Hydraulic actuators are preferred over electric motors when the task requires generating forces of several tonnes.

Answer: True

Hydraulic systems have very high force density and can generate forces far beyond comparably sized electric motors.

9. A stepper motor will automatically send a fault signal to the controller if it skips a step due to excessive load.

Answer: False

Stepper motors have no position feedback by default. A skipped step goes completely undetected by the controller.

10. A LiDAR sensor can operate effectively in complete darkness because it generates its own laser pulses and does not need ambient light.

Answer: True

LiDAR is an active sensor that emits its own light and measures the return signal making it fully independent of ambient lighting.

11. Sensor fusion can help a robot detect a glass door that LiDAR misses provided another sensor such as a camera can observe the door frame.

Answer: True

When one sensor fails in a particular scenario another can compensate. Fusion combines the strengths of multiple sensor modalities to cover individual weaknesses.

12. A robot relying only on wheel odometry will maintain accurate localisation over a 200-metre journey even on a smooth floor.

Answer: False

Small errors per step accumulate through integration over long distances. Even on a smooth floor odometric drift becomes significant over 200 metres.

13. A belt drive and a chain drive used in the same robot will perform identically in all operating conditions including wet cold and high-load environments.

Answer: False

Belts are quieter and smoother but degrade in wet cold and high-load conditions. Chains are noisier but more durable in those conditions. They are not equivalent across all environments.

14. A rack-and-pinion mechanism converts rotary motion from a motor into linear motion along a track.

Answer: True

This is the fundamental function of a rack-and-pinion system. The rotating pinion gear drives linear motion of the rack.

15. A PLC is more suitable than a Raspberry Pi for controlling a welding robot that must operate 24 hours per day with zero tolerance for software crashes or timing variation.

Answer: True

PLCs are hardened for continuous industrial operation and deterministic execution. Linux SBCs can experience OS-level interruptions unacceptable in safety-critical industrial applications.

16. A DC motor without position feedback is an ideal choice for any application that requires the joint to return to exactly the same angle on every cycle.

Answer: False

Without feedback the controller cannot know the actual position or correct errors. Consistent precise positioning requires a feedback mechanism such as an encoder.

17. The reactive layer in a hybrid robot can respond to a sudden obstacle before the planning layer has finished computing a new path.

Answer: True

The reactive layer operates independently and immediately from sensor input. It does not wait for the planning layer to complete a new plan.

18. An infrared sensor will perform equally well detecting a shiny white surface and a matte black surface at the same distance under identical lighting.

Answer: False

Shiny white surfaces reflect infrared well. Matte black surfaces absorb most of the infrared energy returning very little to the receiver and causing weaker or absent readings.

19. A pneumatic actuator is the best choice for a surgical robot that must make sub-millimetre position corrections repeatedly.

Answer: False

Compressed air is compressible and reacts to varying loads by changing volume making sub-millimetre precision very difficult without highly complex compensation. Electric servo actuators are far better suited.

20. An encoder mounted on the motor shaft will detect positional errors introduced by mechanical play or backlash in the transmission between the motor and the joint output.

Answer: False

The encoder measures motor shaft rotation only. Any backlash or slip in the transmission between the motor and joint is invisible to the motor-side encoder introducing undetected error at the joint.

21. A robot using only a hierarchical control paradigm can react instantly to a child who runs into its path during a planned corridor traversal.

Answer: False

Hierarchical control must complete or interrupt the current plan cycle re-sense and re-plan before responding. This delay makes instant reaction to sudden events impossible.

22. LiDAR performs poorly in heavy rain because raindrops reflect laser pulses and create many false obstacle readings throughout the scan.

Answer: True

Raindrops are physical objects that each return a laser pulse filling the point cloud with false positives.

23. A stepper motor controller that loses track of position due to mechanical overload will automatically re-synchronise to the correct position when the overload is removed.

Answer: False

Once steps are missed the controller continues from the wrong assumed position. A homing routine moving to a physical reference is required to restore accurate position knowledge.

24. An Arduino Uno is capable of running real-time SLAM using a depth camera at 30 frames per second.

Answer: False

SLAM at 30 FPS requires millions of floating-point operations per frame and substantial RAM. The Arduino Uno has 2 KB of RAM and no floating-point unit making this completely infeasible.

25. The Four Ds framework suggests that tasks which are dangerous for humans are strong candidates for robotic automation regardless of whether the task is technically simple.

Answer: True

Danger to human life is a primary criterion in the Four Ds. If a task places humans at serious physical risk automation is justified even when the task itself is technically straightforward.

26. A belt drive is more suitable than a chain drive for a robot that must operate quietly in a library book-shelving environment.

Answer: True

Belt drives are significantly quieter than chain drives because they avoid metal-on-metal contact making them appropriate for noise-sensitive environments.

27. In a differential-drive robot setting both wheels to the same speed in opposite directions causes the robot to rotate in place around its centre point.

Answer: True

Equal and opposite wheel speeds create a turning moment with zero net forward translation causing the robot to spin in place.

28. A robot navigating a long symmetric corridor using only a camera will always know exactly which door it is in front of because cameras provide rich visual information.

Answer: False

If all doors look identical the camera returns the same observation at every door. This perceptual aliasing makes unambiguous localisation impossible from camera alone.

29. When a gear system reduces output speed relative to input speed the output torque increases proportionally.

Answer: True

This is the fundamental principle of gear reduction. Speed and torque trade off inversely assuming no losses.

30. A robot operating in a completely GPS-denied underground environment can use LiDAR combined with an IMU to estimate its position and build a map simultaneously.

Answer: True

LiDAR-based SLAM combined with IMU motion estimation enables localisation and mapping in GPS-denied environments such as tunnels and underground spaces.

31. A servo motor and a stepper motor will perform identically when a joint must resist an external load and maintain exact position under that load.

Answer: False

A servo motor continuously measures and corrects position under load. A stepper motor has no feedback and will lose position silently if the load exceeds its holding torque.

32. A robot using a purely reactive paradigm will always find the optimal shortest path through a maze because it reacts immediately to sensor information.

Answer: False

Reactive systems have no global path knowledge. They can get trapped in local minima and may never reach the goal even when one exists.

33. A chain drive requires more regular maintenance than a belt drive because chains need lubrication and tension adjustment over time.

Answer: True

Metal chains require periodic lubrication to prevent rust and wear and tension adjustment as the chain stretches. Belts require less ongoing maintenance though they eventually need replacement.

34. A mobile robot can correct odometric drift by recognising a known landmark in the environment and updating its position estimate based on that observation.

Answer: True

Landmark-based correction is a standard localisation technique. When the robot recognises a known feature it can correct accumulated drift in its position estimate.

35. A pneumatic actuator responds faster than a hydraulic actuator for the same applied pressure because air is lighter than oil.

Answer: True

Air has much lower density and viscosity than hydraulic oil. Pneumatic systems can respond faster for light to medium loads though they cannot match hydraulics for heavy-duty force generation.

36. A robot arm using a large output gear driven by a small input gear will have higher output torque but lower output speed compared to using equal-sized gears.

Answer: True

A larger output gear reduces speed and multiplies torque. This is the basic principle of gear reduction.

37. An ultrasonic sensor is well suited to detecting thin metal rods of 3 mm diameter at a distance of 4 metres in open air.

Answer: False

Very thin objects return only a tiny fraction of ultrasonic energy. At 4 metres the echo from a 3 mm rod is likely too weak to produce a reliable detection.

38. Asimov's Third Law allows a robot to protect itself even if doing so would conflict with the Second Law requiring obedience to a human.

Answer: False

The Third Law requires a robot to protect its own existence only when doing so does not conflict with the First or Second Laws. Self-protection cannot override obedience to a human. The Second Law takes priority over the Third Law.

39. A hybrid robot always re-plans its entire global path every time the reactive layer detects an obstacle.

Answer: False

In a hybrid system the reactive layer handles local obstacles immediately without triggering a full global re-plan. Only if the obstacle permanently blocks the planned route would the planning layer be asked to re-plan.

40. An RGB camera can detect and identify a person hidden behind a thin wooden door in a dark room.

Answer: False

An RGB camera cannot see through opaque surfaces and requires ambient light to form an image. A thermal camera could detect body heat but a standard camera cannot.

41. A rack-and-pinion mechanism with a smaller pinion gear will move the rack a shorter distance per motor revolution than the same mechanism with a larger pinion gear.

Answer: True

A smaller pinion has a smaller pitch circumference. Each revolution travels a shorter distance along the rack.

42. A robot performing repetitive screwing tasks on an assembly line needs high repeatability more than high absolute accuracy.

Answer: True

Repeatability means the robot returns to the same position consistently each cycle. For a fixed repetitive task consistency matters more than the ability to reach any arbitrary point in space precisely.

43. A hydraulic actuator and a pneumatic actuator both use fluid pressure to generate motion but hydraulic systems use oil while pneumatic systems use compressed air.

Answer: True

This is the fundamental physical difference between the two systems. Oil is incompressible and enables high force. Air is compressible and suited to faster lighter applications.

44. A robot whose planning layer has an outdated map may plan a route that no longer exists and the reactive layer must handle the discrepancy when the robot encounters the actual change.

Answer: True

The planning layer operates on its stored map. If the map is outdated the planned path may be invalid. The reactive layer detects unexpected obstacles and responds locally.

45. A single-board computer such as a Raspberry Pi is a better choice than a PLC for a robot that must guarantee a motor command is issued within exactly 1 millisecond of a sensor trigger.

Answer: False

Linux-based SBCs have variable task scheduling. A 1-millisecond deterministic response cannot be guaranteed without a real-time OS or dedicated hardware. A PLC or real-time microcontroller is appropriate.

46. A belt drive can absorb mechanical vibration and shock better than a rigid gear train because the belt material has some natural elasticity.

Answer: True

The elasticity of the belt material dampens shock loads and vibration which can protect the motor and driven components from sudden impact forces.

47. In a robot with hybrid control the robot will stop permanently if the reactive layer detects any obstacle anywhere along the planned route.

Answer: False

The reactive layer avoids the obstacle locally and the robot continues toward the goal. Stopping permanently is the behaviour of poorly designed or purely hierarchical systems.

48. An infrared sensor works well at detecting clear transparent objects such as a glass bottle directly in front of the robot.

Answer: False

Transparent objects transmit infrared rather than reflecting it back to the sensor. A clear glass bottle will be largely invisible to a standard infrared proximity sensor.

49. A robot using only LiDAR will be unable to distinguish between a wooden door and a metal door at the same distance.

Answer: True

Standard LiDAR measures distance from returned laser pulses. It does not measure material properties such as composition or texture. Both surfaces at the same distance look identical in a LiDAR scan.

50. When a robot wheel encoder counts more rotations than the actual ground distance covered due to wheel slip the robot estimated position will be ahead of its true position.

Answer: True

The encoder counts rotations as if they all resulted in forward motion. Slip means some rotations did not produce movement so the estimated position overshoots the real position.

Part C: Descriptive (15 Scenarios)

1. Situation: The Frosted Glass Warehouse

A mobile robot operates in a refrigerated warehouse where walls are made of frosted glass panels and floors are polished concrete. The robot uses an ultrasonic sensor and an infrared sensor for obstacle detection. On a routine run the robot approaches a frosted glass wall at a 40-degree angle and fails to stop in time.

- (a) Identify which sensor most likely failed and explain the physical reason why the frosted glass wall at 40 degrees caused the failure.
 - (b) Recommend a sensor combination to fix this specific problem and justify why each sensor handles the failure case the other missed.
 - (c) The polished concrete floor also causes problems for the IR floor-edge sensor. Describe the failure mode and how the sensor setup should be adjusted.
-

2. Situation: The Silent Factory Arm

An industrial robotic arm in a food packaging factory picks small chocolate pieces from a conveyor and places them into trays at 120 pieces per minute. The arm has six joints and the factory operates 18 hours per day. Workers are nearby so noise is a concern. Downtime costs are very high.

- (a) Select a motor type for the joints and justify based on precision continuous duty cycle and reliability requirements.
 - (b) Select a transmission type and justify based on noise level precision and suitability for a food environment where hygiene is important.
 - (c) Select control hardware and justify based on reliability uptime and deterministic cycle control over an 18-hour shift.
-

3. Situation: The Stairwell Inspection Robot

A robot is deployed to autonomously inspect stairwells in a multi-storey building for structural cracks and damage. Stairs are narrow and lighting varies floor to floor. The robot must navigate without a pre-built map. Workers may appear unpredictably on the stairs.

- (a) Choose a control paradigm for this robot and justify your choice given the unstructured and dynamic environment.
 - (b) Recommend a sensor set for navigation crack detection and worker avoidance. Justify each sensor role in this specific stairwell environment.
 - (c) Explain why a purely hierarchical control approach would be particularly dangerous in a stairwell environment with unpredictable workers.
-

4. Situation: The Underground Cable Robot

A small robot crawls through underground cable ducts that are 40 cm in diameter completely dark and often dusty with fine fibreglass particles. It must map the duct interior and detect blockages or cable damage. No GPS signal is available.

- (a) Recommend sensors for mapping and blockage detection in a dark dusty confined environment. Explain why outdoor navigation sensors may fail here.
- (b) Choose an actuation approach for mobility through the duct and justify for confined low-clearance conditions.

(c) Without GPS describe the localisation strategy the robot should use and identify the main source of error in that strategy over long duct runs.

5. Situation: The Orchard Harvesting Robot

A robot picks ripe apples from trees in an outdoor orchard. Fruit hangs at varying heights. Lighting changes from bright sun to full shade as the robot moves between trees. The arm must grasp apples without bruising them. The soil is soft and uneven.

- (a) Choose a control paradigm and justify for the combination of planned row-following routes and unplanned individual fruit positions within those rows.
 - (b) Select motor type and gripper actuation for gentle grasping and justify why excessive grip force must be actively prevented by the control system.
 - (c) Identify two sensors that would fail under bright direct sunlight and explain the physical reason for each failure. Recommend what should replace or supplement them.
-

6. Situation: The Museum Night Watchman Robot

A robot patrols a museum at night checking for intruders fire and fallen artefacts. The museum has large rooms with exhibits of complex shapes. Lights are dimmed. The robot must not touch or disturb any exhibit. It must send alerts immediately if anything is wrong.

- (a) Choose a control paradigm and justify for autonomous patrol with real-time threat detection.
 - (b) Choose sensors for intruder detection fire detection and artefact position monitoring. Justify each and explain why a camera alone is insufficient at night.
 - (c) The path-planning layer assumes the museum layout is fixed. A cleaner has moved a display stand overnight. Describe what happens at the planning layer and the reactive layer respectively when the robot encounters the moved stand.
-

7. Situation: The Bridge Inspection Drone

A UAV inspects the underside of a large suspension bridge over a river. The bridge has steel cables concrete surfaces and complex metallic geometry. Wind gusts are unpredictable. GPS signal is weak under the bridge deck due to signal blockage by the structure.

- (a) Identify two sensors that will have specific failure modes under this bridge environment and explain the physical reason for each failure.
 - (b) Choose an appropriate sensing solution for safe close-range navigation under the bridge and justify why it handles the specific conditions present.
 - (c) The UAV must maintain a 0.5-metre standoff distance from the bridge surface. Explain what type of control behaviour handles this in real time and which paradigm layer it belongs to.
-

8. Situation: The Printing Press Alignment Robot

A robotic arm must continuously align a print head to a target mark on a paper roll moving at high speed. Any overshoot or oscillation in the arm position ruins the printed output. The task runs without stopping for eight hours and precision must remain consistent throughout.

- (a) Choose a motor type and justify based on the need for sub-millimetre precision under continuous operation with varying load from the moving paper.
- (b) Explain why a stepper motor without feedback is inappropriate for this task even though stepper motors can move in small precise increments.

- (c) Choose control hardware and justify based on the need for deterministic timing when the paper is moving continuously and corrections must be issued in real time.
-

9. Situation: The Tilted Solar Panel Cleaning Robot

A robot cleans solar panels on a rooftop installation. Panels are tilted at 30 degrees. The robot must detect soiled areas and apply cleaning fluid without falling off the panel edge. The panel surface is reflective glass and the background beyond the edge is open sky.

- (a) Choose a drive and actuation mechanism and justify for operation on a 30-degree tilted smooth glass surface where grip and controlled movement are both critical.
 - (b) Select sensors to detect the panel edge reliably given that the surface is reflective glass and the region beyond the edge is open sky which may confuse distance sensors.
 - (c) Explain what control paradigm would allow the robot to follow a planned cleaning path while immediately stopping if an edge is detected unexpectedly. Describe what each layer does in this scenario.
-

10. Situation: The Underwater Pipe Inspection Robot

A robot travels through underwater oil pipes of 60 cm inner diameter to detect internal corrosion. Water pressure is high communication to the operator has a significant delay and metallic pipe walls completely surround the robot.

- (a) Identify three sensors commonly used in land robots that will fail in this environment and explain why each one fails.
 - (b) Recommend sensors that can work in this specific underwater enclosed metallic environment and justify each recommendation.
 - (c) With a significant communication delay to the operator explain whether a hierarchical or hybrid control paradigm is more appropriate and why the robot must be capable of making some decisions autonomously.
-

11. Situation: The Swinging Door Navigation Robot

A hospital delivery robot approaches an automatic door that swings outward when triggered and then swings back closed. The robot must pass through safely and must not get caught by the closing door. The corridor is busy with patients and medical staff who may be near the door.

- (a) Explain what sensing approach would allow the robot to detect the door current state and position before deciding whether to proceed through.
 - (b) The reactive layer must decide whether to move through or wait. Describe what observable sensor condition should trigger the move-through decision versus the wait decision.
 - (c) If the door malfunctions and stops halfway open describe how the robot planning layer and reactive layer should each respond to this unexpected situation.
-

12. Situation: The High-Speed Sorting Robot

A robot arm sorts packages on a conveyor belt into three chutes based on package colour and weight. Packages arrive at regular intervals. Some packages are dark-coloured and the environment has variable fluorescent lighting that flickers slightly throughout the day.

- (a) Choose a motor type for the sorting arm and justify based on speed repeatability and the very high number of cycles the arm must complete per shift.

(b) The colour detection sensor fails on dark packages under flickering lighting. Identify the sensor type that has this weakness and recommend a backup or supplementary sensing strategy that is more robust to lighting variation.

(c) The robot must make a chute selection decision within a tight time window. Explain why a microcontroller alone is insufficient if the decision also requires running a classification algorithm on the sensor data and recommend an appropriate control hardware combination for this task.

13. Situation: The Search and Rescue Ground Robot

A ground robot is deployed into a partially collapsed shopping mall after an earthquake. The structure is unstable rooms are filled with dust and debris and survivors may be anywhere. The robot has no prior map and must find survivors and report their locations.

(a) Choose a control paradigm and justify for a robot that must explore unknown dangerous terrain while reacting immediately to structural hazards.

(b) Recommend a sensor set that allows the robot to detect survivors navigate in dust-filled dark spaces and detect structural instability. Justify each sensor choice.

(c) The robot loses communication with the operator for several minutes. Explain what control behaviour should continue autonomously during the communication gap and what should happen when communication is restored.

14. Situation: The Textile Quality Inspection Robot

A robot arm moves along a track above a textile production line. It inspects fabric rolls for weaving defects colour inconsistencies and surface damage. The fabric moves continuously below the arm. The room has strong overhead lighting but the fabric surface can be highly reflective depending on material type.

(a) Select sensors for detecting weaving defects and colour inconsistencies on a moving reflective fabric surface. Justify each sensor suitability and identify any limitation.

(b) Select a transmission for the arm movement along the track and justify based on positional consistency low vibration and the need to not contaminate the fabric with lubrication residue.

(c) The inspection system must trigger an alert within a very short time window when a defect is detected. Choose control hardware and justify based on timing reliability and the need to interface directly with both the sensing system and the conveyor stop mechanism.

15. Situation: The Autonomous Guided Vehicle in a Smart Warehouse

A fleet of autonomous guided vehicles navigates a large smart warehouse. Each vehicle carries loads between storage racks and dispatch stations. Other vehicles human pickers and forklifts all share the same floor space. The warehouse layout changes monthly as new product sections are added.

(a) Choose a control paradigm and justify for a vehicle that must follow assigned delivery routes while safely sharing space with humans forklifts and other robots in real time.

(b) Recommend a sensor set that provides reliable obstacle detection and localisation in an environment with metallic shelving racks variable lighting and frequent layout changes. Justify each sensor specific role.

(c) When the warehouse layout changes the vehicle map becomes outdated. Describe how the system should handle this: what happens at the planning layer when the vehicle encounters a rack that is no longer there and what happens when a new obstacle appears where the map shows open space.
